WitMotion M10 in OpenMower

Communication and RTK diagnostic guide | English | 20 October 2026

For a WitMotion / ByNav WTRTK-M10 used as the rover receiver in an OpenMower robot. Start with non-destructive tests. Configuration changes are isolated at the end. This is not a guide for a u-blox M10 or a ByNav M20.

Privacy: no personal IP addresses, coordinates, credentials, serial numbers or original logs are included. ROBOT_IP, BASE_IP and ROBOT_USER are placeholders. Replace them locally. 127.0.0.1 is the standard loopback address, not a personal address.

Before testing: park the robot, stop mowing and keep the blade disabled. Do not reset the receiver, change its baud rate or open several serial clients at once during the initial checks. Satellite reception is not required for a receiver to answer a command.

Where commands run

PC: your computer terminal. **HOST:** the robot Raspberry Pi SSH shell. **ROS:** the shell opened by `openmower shell`. **M10:** ASCII commands sent to the receiver; these are not Linux shell commands.

```
# PC: replace both placeholders
ssh ROBOT_USER@ROBOT_IP

# HOST: enter the ROS container
openmower shell

# ROS: return to the HOST
exit
```

Diagnostic order: 1 software health; 2 ROS data; 3 direct receiver access; 4 command replies; 5 optional USB isolation; 6 correction reception; 7 restore and confirm FIX; 8 configuration repair only if necessary.

Key distinction: a working network connection does not prove communication with the M10. A valid M10 response proves communication. A position proves GNSS reception. NARROW_INT / RTK FIX additionally proves an integer RTK solution at that instant.

1. Check the robot software

HOST - read-only commands

```
openmower status
command -v python3
sudo docker ps --format 'table {{.Names}}■{{.Status}}'
sudo docker logs --since 10m --tail 500 open_mower_ros 2>&1 \
  | grep -Ei 'gps|gnss|rtcm|ntrip|serial|uart|error|warn' \
  | tail -n 80
```

Expected: the ROS container is running; Python 3 is available for the diagnostic scripts. Use the actual ROS container name if it differs from open_mower_ros. A running container alone does not prove that GPS communication works.

Interpretation: repeated serial, connection or parsing errors point toward the receiver link or its configuration. “Dropped GPS update, since it is not RTK Fixed” indicates a solution-quality filter; it is not proof of a disconnected receiver.

ROS - discover the actual topics and configuration names

```
openmower shell
rostopic list | grep -Ei 'gps|gnss|ntrip|rtcm|diagnostic'
rosparam list | grep -Ei 'gps|gnss|baud|protocol'
rostopic info /ll/position/gps
rostopic type /ll/position/gps | xargs rosmsg show
```

Run `rosparam get EXACT_PARAMETER_NAME` for the GPS baud rate and protocol names returned above. Do not publish a complete parameter dump: it may contain NTRIP credentials.

Expected on the tested v2 layout: /ll/position/gps, /ll/position/gps/nmea and /ll/position/gps/rtcm. If a topic is absent, inspect the topics actually listed; do not assume /gps exists. Different OpenMower versions can use different names.

Important: a carrier-board M10 need not appear as /dev/ttyUSB0 on the Raspberry Pi. OpenMower can access it through the controller and a TCP-to-serial bridge. USB discovery is relevant only when the receiver is physically connected by USB.

2. Check what OpenMower receives

ROS - bounded tests; timeout termination is expected

```
timeout 10 rostopic echo -n 2 /ll/position/gps
timeout 15 rostopic hz /ll/position/gps
timeout 10 rostopic echo -n 20 /ll/position/gps/flags
timeout 15 rostopic echo -n 2 /ll/position/gps/nmea
```

Expected: messages arrive with increasing sequence numbers or timestamps. A typical configured receiver updates at about 10 Hz, but the configured rate and software version determine the actual rate.

Flags: first verify the constants printed by rosmsg show. In the tested message definition, RTK=1, RTK_FIXED=2 and RTK_FLOAT=4. Therefore 3 means RTK + FIXED and 5 means RTK + FLOAT. These are bit flags: other bits may also be present.

position_accuracy: an estimated accuracy in metres in this implementation, not a surveyed absolute error. A small value with FLOAT must not be interpreted as FIX. Local pose x/y values are not latitude and longitude.

NMEA warning: the ROS NMEA topic can contain a synthesized GGA sentence for the NTRIP client. Placeholder quality or satellite fields there may not represent the receiver. Use direct M10 output in the next steps to establish the true GNSS state.

If there are no messages: check the topic publisher, logs and direct receiver access. If the module reports no position but responds to commands, communication already works; investigate antenna/reception separately.

If messages arrive: save only the relevant, anonymized fields for comparison. Do not change GPS settings merely because the receiver is FLOAT. Proceed to the correction tests after verifying direct replies.

3. Open a temporary direct GPS connection

HOST, terminal A: leave this command running. If inside ROS, type exit first.

```
openmower expose-gps
```

Expected: a message saying that a TCP connection is available on port 2000, followed by the bridge startup. The displayed internal controller address is installation-specific; do not copy an address from someone else.

HOST, terminal B: open a second SSH session to the robot, then check the listener.

```
sudo ss -ltnp | grep ':2000'
python3 - <<'PY'
import socket, time
with socket.create_connection(('127.0.0.1', 2000), timeout=5) as s:
    s.settimeout(1)
    end = time.monotonic() + 10
    total = 0
    sample = bytearray()
    while time.monotonic() < end:
        try:
            b = s.recv(8192)
        except socket.timeout:
            continue
        if not b:
            break
        total += len(b)
        if len(sample) < 300:
            sample.extend(b[:300-len(sample)])
    print('Bytes received:', total)
    print('First bytes:', repr(bytes(sample)))
PY
```

Expected: readable $G... sentences or #... messages if periodic output is enabled. Zero bytes does not prove a failed receiver: it might have no periodic logs enabled. Continue with a query.

Connection refused: the expose service is not listening. **TCP connects but no replies:** the bridge can exist while the hardware link is broken. **Garbled bytes:** investigate baud rate and framing; some output can also be legitimate binary data.

Possible side effect: direct access may interfere with normal GPS or correction delivery. Do not assess long-term RTK stability while this bridge is open. Close it as described in step 7.

4. Query the M10 without reprogramming it

HOST, terminal B: keep expose-gps running in A. This script requests one-shot reports with CR/LF endings. It does not reset or save configuration. Coordinates may appear in the results: redact them before sharing.

```
python3 - <<'PY'
import socket, time, re
commands = ['LOG VERSION ONCE', 'LOG COMCONFIG ONCE',
            'LOG BESTPOSA ONCE', 'LOG LOGLISTA ONCE']
with socket.create_connection(('127.0.0.1', 2000), timeout=5) as s:
    s.settimeout(0.3)
    for command in commands:
        print('\nREQUEST:', command, flush=True)
        s.sendall((command + '\r\n').encode('ascii'))
        end = time.monotonic() + 4
        buffer = b''
        shown_gga = False
        found = False
        while time.monotonic() < end:
            try:
                chunk = s.recv(8192)
            except socket.timeout:
                continue
            if not chunk:
                break
            buffer = (buffer + chunk)[-65536:]
            while b'\n' in buffer:
                line, buffer = buffer.split(b'\n', 1)
                line = line.rstrip(b'\r')
                if not line or any(c < 32 or c > 126 for c in line):
                    continue
                text = line.decode('ascii')
                if re.match(r'^\$[A-Z]{2}GGA,', text):
                    if not shown_gga:
                        print(text)
                        shown_gga = True
                elif text.startswith(('#', '<', '$BDVER')):
                    print(text)
                    found = True
        if not found:
            print('No matching reply captured; not proof of failure.')
print('Finished. No persistent setting changed.')
PY
```

VERSION: firmware/identity. **COMCONFIG:** receiver port settings. **BESTPOSA:** solution type, satellites, uncertainty and correction age. **LOGLISTA:** enabled periodic outputs and their internal COM ports.

An explicit OK or structured report proves that commands reach the module. An unknown-command error still proves communication; syntax can vary with firmware. A filtered script may miss a reply mixed with binary data. Bare RTKTYPE / RTKTIMEOUT queries did not return captured values in the previous investigation and should not be used as the only communication test.

5. Optional USB isolation test

Use this only if direct robot access fails. Power the robot down before removing the module. Connect the M10 to a Linux computer using a known data-capable USB cable. Avoid powering it simultaneously through the carrier and USB unless the hardware documentation allows it.

```
lsusb
ls -l /dev/serial/by-id/
sudo dmesg | tail -n 40
ls -l /dev/ttyUSB0
id
sudo fuser -v /dev/ttyUSB0
command -v picocom
```

Expected: a USB-to-serial adapter and a serial device. CH340 / 1a86:7523 was seen on the investigated board; variants can differ. Replace ttyUSB0 with the discovered device. A USB adapter appearing proves USB enumeration, not communication with the GNSS chip.

If another program owns the port, identify it before stopping it. On a separate RTKBase Pi, str2str normally owns the receiver. Do not let picocom and str2str read the same port simultaneously.

If picocom is already installed:

```
picocom --baud 115200 --flow n /dev/ttyUSB0
```

For a permission error, verify membership of the device group (often dialout); use sudo picocom for a one-off test if appropriate. Do not chmod the device to world-writable. Exit picocom using Ctrl+A, then Ctrl+X. If unavailable, use an existing serial terminal such as the manufacturer tool; installation is not required for the TCP tests.

In the serial terminal, send CR/LF-terminated M10 commands:

```
LOG VERSION ONCE
LOG COMCONFIG ONCE
LOG BESTPOSA ONCE
LOG LOGLISTA ONCE
```

115200 is the documented starting USB rate. A previously configured port may differ. If unreadable, close the terminal and retry the known configured rate. For an unknown unit, try 115200 then 460800 without issuing reconfiguration commands. Do not confuse USB COM3 with the carrier connection on COM1.

Isolation result: USB replies but carrier access does not: inspect COM1 rate, carrier contacts/header soldering and robot configuration. No USB enumeration: investigate cable, power or adapter. Adapter present but no replies at plausible rates: verify framing, firmware command syntax and module power before concluding hardware failure.

6. Verify corrections reaching the robot

Close picocom or the direct TCP client and stop expose-gps with Ctrl+C in terminal A before this test. These checks separate the network from the receiver.

HOST: replace BASE_IP with the actual caster address; 2101 is a common port, not a requirement.

```
ss -tn | grep 'BASE_IP:2101'
openmower shell
```

ROS:

```
timeout 10 rostopic echo -n 1 /ntrip_status
timeout 10 rostopic echo -n 1 /ntrip_status_code
timeout 15 rostopic echo -n 3 /ntrip_counters
rostopic info /ll/position/gps/rtcm
timeout 15 rostopic hz /ll/position/gps/rtcm
```

Expected: established TCP connection, active stream, increasing bytes_received and frames_published, no increasing CRC failures. The tested wiring had /ntrip_client as publisher and /mower_comms_v2 as subscriber. Different versions may name nodes differently.

RTCM frames arrive in bursts. Several frames per second are normal; the initial hz estimate can be misleading. This frequency is a frame rate, not the GNSS measurement rate. A short “no new messages” line needs context, not an immediate configuration change.

What this proves: correction data reaches ROS. It does not alone prove that the M10 receives or uses those corrections. Direct BESTPOSA correction age and solution type provide the next level of evidence.

If counters stay still: check caster address, port, mountpoint, credentials and Wi-Fi reachability. If counters increase but there is no subscriber, inspect the robot communication node. If reception works but FLOAT persists, inspect common satellite signals, antenna placement and base observations rather than repeatedly changing network settings.

In the earlier investigation, missing GPS and BeiDou output on the base was confirmed by its active log list. Enabling the corresponding base messages restored populated RTCM 1077 and 1127. Valid CRCs alone had not revealed that missing measurement content. Do not apply base logging commands to the robot.

7. Restore normal operation and judge the result

First: close the Python/serial client, then Ctrl+C the expose-gps process. Return to normal OpenMower operation. If it does not resume, use the normal restart command while the robot is parked:

```
# HOST: only if normal operation has not resumed
openmower restart

# HOST: enter ROS for verification
openmower shell
timeout 15 rostopic echo -n 3 /ntrip_counters
timeout 10 rostopic echo -n 2 /ll/position/gps
timeout 60 rostopic echo /ll/position/gps/flags
```

Expected: live timestamps, growing correction counters and stable FIX flags when reception conditions allow. Repeated 3 values in the tested message definition mean RTK FIX. Confirm stability over several minutes outdoors, not just one message.

True receiver fields: direct GGA quality 0 means invalid, 1 standalone, 4 RTK fixed and 5 RTK float. BESTPOSA SINGLE means autonomous, NARROW_FLOAT means float and NARROW_INT means fixed. Base-specific fixed-position modes are not the same as a rover resolving RTK ambiguities.

Antenna placement matters: both antennas need a clear sky view; walls and nearby reflective structures can damage carrier-phase measurements even with many satellites. A receiver indoors may communicate perfectly but fail to obtain a useful RTK solution.

Observed outcome in the previous investigation: after restoring missing base constellation outputs and improving placement, the robot reported stable flags=3 and direct NARROW_INT. The reported accuracy was approximately 0.014 m. During a direct diagnostic connection, correction age rose and the solution fell to SINGLE; normal FIX returned after closing that connection. The interruption mechanism was not conclusively established.

Do not equate FIX with correct mapping: moving the base after fixing its coordinates can shift the robot coordinates. A centimetre-scale precision estimate does not validate the absolute base position. Keep the base antenna stable and reconcile its configured reference before recording boundaries.

8. Optional rover configuration repair

Changes settings. Only do this after communication works and the current settings have been recorded with COMCONFIG and LOGLISTA. Never send FIX POSITION to the robot. Do not factory-reset a working receiver as a diagnostic shortcut.

For the WitMotion / ByNav M10, OpenMower documents COM1 as the carrier interface. The following is the documented rover profile. Send these in the M10 terminal, not at the Linux prompt. USB COM3 is the easier route because changing COM1 need not disconnect USB.

```
SERIALCONFIG COM1 460800
RTKTYPE ROVER
RTKTIMEOUT 5
LOG COM1 GPGSV ONTIME 1
LOG COM1 GPRMC ONTIME 1
LOG COM1 GPGSA ONTIME 1
LOG COM1 GPVTG ONTIME 1
LOG COM1 GPGST ONTIME 1
LOG COM1 GPGGA ONTIME 0.1
LOG COMCONFIG ONCE
LOG LOGLISTA ONCE
SAVECONFIG
```

Critical: over TCP/COM1, the first command changes the rate immediately. Stop the bridge, update the robot baud rate, then reconnect before sending the remaining commands. Over USB/COM3, verify its own rate rather than changing it to match COM1.

```
# HOST: edit the existing GPS section, preserving other settings
openmower configure ros

gps:
  baud_rate: 460800
  protocol: "NMEA"
```

Merge these keys into the existing configuration at the correct level; do not create a duplicate gps section. Apply with openmower restart while parked, then repeat steps 2-7. Review command replies before SAVECONFIG: it persists changes across power cycles.

Commands deliberately excluded from routine tests: FRESET resets the receiver and may change port rates. UNLOGALL stops cyclic output; it can make a working receiver appear silent. Both occur in the official clean-install workflow, but neither is necessary to prove communication. Use that workflow only when deliberately rebuilding the configuration.

9. Results checklist and references

Record the first failing stage:

• Robot software: container running; GPS topic publisher present.

• Direct bridge: port 2000 listening; TCP connection succeeds.

• Receiver communication: explicit reply to VERSION or another query.

• Port settings: actual carrier COM1 rate matches the robot configuration.

• Receiver output: COM1 contains the expected NMEA logs.

• GNSS reception: valid standalone or RTK position outdoors.

• Corrections: counters grow; CRC failures do not grow; subscriber exists.

• Receiver correction use: direct reports show a plausible correction age.

• RTK solution: NARROW_INT / quality 4 and corresponding ROS FIX flags.

• Normal operation: direct access closed; FIX remains stable.

When asking for help: include the OpenMower version/image tag, module model, interface used, configured baud rate, exact command and error, and the checklist stage that fails. Remove coordinates, public/private network addresses, passwords, keys, receiver serial numbers and identifying hostnames from logs.

Official documentation reviewed

OpenMower - Prepare the GPS (WitMotion / ByNav M10 configuration):
openmower.de - Prepare the GPS

OpenMower - Configure ROS:
openmower.de - Configure ROS

The diagnostic scripts and command sequence in this guide were assembled for this troubleshooting workflow. The earlier outcomes are anonymized observations, not guaranteed results for another installation. Topic names, tools and supported receiver queries can differ by software and firmware version.